An Interactive User-Centric Airline Recommendation System

Fall 2023, CSE 6242 Data & Visual Analytics, Team045 (5 members)

Amber Wo (jwo6), Yi Liu (yliu330), Liulinbo Grace (lgrace7), Qi Pan (qpan40), Zhaowei Meng (zmeng65)

1 INTRODUCTION

Many travelers today choose flights based on price and schedule, leaving uncertainty about service quality. This approach could potentially lead to varied and sometimes unpleasant travel experiences. Our project aims to bridge this gap by utilizing user feedback and employing an advanced algorithm to provide personalized airline suggestions that align with travelers' individual preferences, providing a more tailored user-centric air travel experience.

2 PROBLEM DEFINITION

It is often a challenge for travelers to select from many airline options for a satisfying flight. Our project introduces an innovative web application that refines the airline selection process and emphasizes travelers' experiences. Leveraging real user data, our system delivers customized airline suggestions, taking into account individual preferences in areas such as food quality, seat comfort, entertainment, cabin service, ground service, and wifi connectivity. This user-centric approach not only significantly enhances the travel planning experience, but also introduces a novel perspective for travelers in choosing airline services. Additionally, it has the potential to serve as a valuable tool for airline companies, providing insights for continuous service improvement and strategic business development.

3 LITERATURE SURVEY

The rise of airline recommendation apps has brought both challenges and advancements. Dadoun *et al.* identified the barrier to personalized airline recommendations arising from a lack of firsthand user reviews [6]. With an increase in social media use, researchers like Heidari *et al.* are exploring these platforms for data, but sentiment classification remains problematic even with their "bag of words" approach [10]. Brendel *et al.* emphasized data pre-processing, addressing biases like missing data [4].

While Cao *et al.* employed the tensor factorization method for personalized flight recommendations using historical data, concerns about model reliability emerged due to varying performance [5]. Thirumuruganathan *et al.* pointed out a gap between model accuracy in theory and real-world application, underscoring the dynamic nature of user preferences [15]. This aligns with Abdollahpouri *et al.*'s insights on addressing algorithmic popularity bias to match varied user preferences [1]. Furthermore, recommendation systems, as dissected by Taghavi *et al.*, exhibit complex layered structures akin to well-known platforms like Facebook and Google [14].

Dimensionality reduction, as explored by Wang and Carreira-Perpiñán, remains pivotal for data visualization, with their novel "wrapper" approach promising better results [16]. On the modeling front, Maesschalck *et al.* drew connections between distance metrics, laying the groundwork for enhanced recommendations [7]. The efficacy of classification models, like the KNN algorithm, in this realm is further emphasized by J. Bobadilla et al.'s survey [3].

Visualization remains a crucial component of the user's experience. Our project, drawing from Harasymiv's work, seeks to offer dynamic flight projections on a global scale [9]. Insights from Eberhardt and Meek will guide our data rendering and handling strategies [8, 12]. Ultimately, influenced by Ashley-Dejo's work, we aim to combine these techniques to offer an enriched user experience [2].

We foresee our project driving future studies, with airlines potentially gaining from user feedback to boost their offerings and growth—a sentiment echoed by Siering *et al.* and Manganari [11, 13].

4 PROPOSED METHOD

4.1 Intuition

Our project provides airline recommendations with features designed for enhanced user engagement and personalized accuracy in an innovative way.

Dynamic Geovisualization and Service Experience Focus. The interface features dynamic geovisualization, triggered by users' selections of departure, connecting, and arrival cities. It offers a real-time, interactive

display that indicates selected city names and enables users to see a virtual flight visualization on the world map. This is also coupled with a Random Forest model that prioritizes service experience based on real user data to accurately align with individual preferences on the importance of service areas, such as seat comfort or in-flight amenities.

Gauge Charts and Airline Exploration. Gauge charts are introduced for intuitive compatibility assessments of airline suggestions, simplifying user decision-making. Additionally, each airline recommendation links to more detailed information about the airline, providing insights into each airline's ratings on various service areas, user reviews, and a link that connects with the airline's official website, ensuring users make informed decisions.

Foundation for Future Enhancements. Our design includes a review section, inviting travelers to share their experiences. This feature not only enriches our database but also aids in enhancing the performance of our model. The current model relies on the Random Forest algorithm, which is robust but shows increased training times as the dataset grows. Future iterations of the system should explore alternative modeling approaches to address this scalability challenge. Such exploration is aimed at reducing the time dependencies observed with the current model and promises further refinement and optimization of our analytical methodologies.

In essence, our approach sets a new benchmark in travel planning tools, blending interactive visualization with a deep understanding of user preferences and the anticipation of evolving needs.

4.2 User Interactive Visualization

User interactive visualization is critical to this project. The front-end development utilizes a combination of HTML, CSS, JavaScript, and D3.js. As a JavaScript library, D3.js plays a crucial role in creating dynamic and data-driven visualizations, such as a seamless world map and gauge charts that display backend airline recommendation data. The core JavaScript enables interactive features, including user preference inputs, review submissions, data loading for city name dropdown menus, and real-time content updates in response to user interactions. CSS is employed for its styling, layout design, and responsive elements, ensuring a visually appealing user interface. The integration of D3.js, HTML, CSS, and JavaScript in this project offers a user-friendly visualization interface.

4.2.1 Homepage Design. As shown in Figure 1, the homepage is structured with a dual-section layout: the control panel and the map visualization, with a 1:4 ratio to optimize readability.

Control Panel. The control panel contains two sections: the menu at the top and the user preference selection area, providing users with easy access to the following functionalities:

- Personalized airline recommendation
- Flight experience review submission

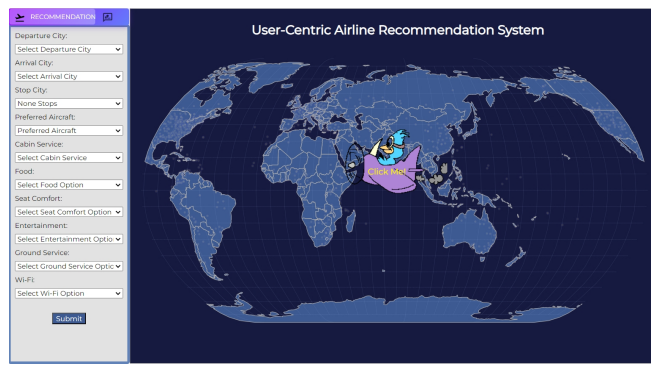


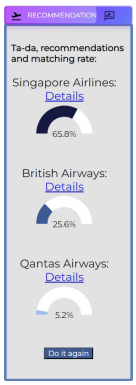
Figure 1: Homepage Visualization

System Introduction. Upon clicking the "Click Me!" button on the cartoon character flying in the middle of the world map, the user interface pops up a window with a brief system introduction to ensure users can easily understand the system's main features, including airline recommendation and review submission, to provide users a smooth and efficient experience.

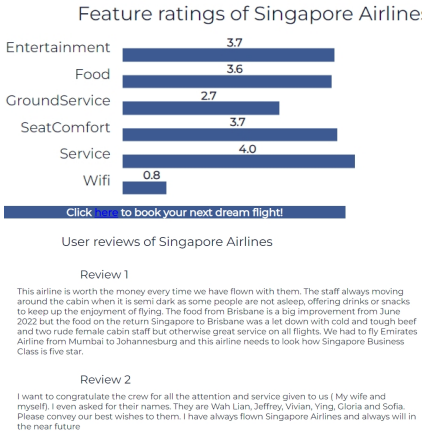
4.2.2 Personalized Recommendation and Airline Exploration. The personalized recommendation is a two-step process: it begins with generating airline suggestions based on user-input preferences on multiple areas of airline services, followed by an airline exploration phase where users have the opportunity to delve deeper into the details of the recommended airlines.

Airline Recommendation Interactive Visualization. Upon selecting departure, stopover, and arrival cities in the control panel on the left side of the application, the

map dynamically spins the world map to center on the selected city and displays its name. A bright blue curved line, accompanied by an airplane icon, connects the cities, visualizing the user's journey. Following the submission of the user's service preference, recommendation results are presented in gauge charts, providing an intuitive visual representation of the recommendation matching, as shown in Figure 2a.



(a) Recommendation



(b) Airline Exploration

Recommended Airline Exploration. By clicking the "Details" link next to each recommended airline, users are guided to the respective airline's feature page for further exploration. As shown in Figure 2b, this feature page includes the airline's service ratings and users' comments, offering users a comprehensive overview.

4.2.3 Review Submission Interface. Under the "Review" section, users can submit their flight experience reviews. Upon submission, a thank you note displays in a pop-up window, acknowledging the user's contribution.

4.3 Recommendation Modeling

At the heart of our recommendation system lies a tailored model based on an authentic trip dataset, highlighting cities, airlines, aircraft types, and service ratings.

User-Defined Metric Incorporation. The model is predicated on user-defined metrics, meticulously chosen to reflect the most impactful aspects of airline service as identified by travelers, such as food, seat comfort, and various service quality. By encoding these preferences into our algorithm, we ensure the model's outputs are tightly coupled with user satisfaction criteria.

Data Modeling and Integration. Data modeling encountered the usual suspects of missing data and the need for normalization. For numeric variables, mean imputation was applied to address the missing data. In addition, we performed comprehensive data cleansing and employed one-hot encoding for categorical variables, ensuring model readiness. The introduction of supplementary data was approached cautiously to maintain the integrity and dimensional consistency of the original dataset.

Algorithmic Strategy and Experience. During our algorithm selection process, various models were evaluated for their accuracy and response time, as these are the primary concerns of our users. Despite Principal Component Analysis (PCA)'s potential for feature reduction, it was excluded due to its minimal impact on performance, leading to a decision that favored a full-spectrum feature approach. The Random Forest (RF) Classifier stood out for its accuracy and proficiency in generating top-3 airline suggestions. Alternative methods, like K-Nearest Neighbors (KNN) presented iteration challenges, and while Logistic regression had its merits, the Random Forest outperformed consistently in accuracy and maintained a comparable response time.

Validation and Parameter Optimization. Cross-validation techniques were employed for K-Nearest Neighbors (KNN), Logistic Regression, and Radial Basis Function (RBF) Kernel Logistic Regression. For Random Forest, however, the use of its Out-of-Bag (OOB) accuracy, which estimates model performance using data not involved in the training of individual trees, eliminated the need for separate cross-validation. This approach provides a comparable assessment of model accuracy and aids in the optimization of parameters.

Integrating Model and User Interface. The RF model's deployment was carefully synchronized with the interactive user interface. The entire process begins when a user inputs their airline service preferences through the frontend user interface. These individual preferences are then transmitted to the backend, where they serve as parameters to filter and select the data for RF model training, ensuring that the training process is optimized to reflect the specific user preferences. As a result, the optimized RF model is constructed and it promptly generates and outputs the top three airline recommendations that match the user's preferences. This integration

allows for swift model response based on user interactions, maintaining the application's agility and user engagement.

4.4 Web Application Architecture

Figure 3 illustrates the flow of requests and responses within our web application. When users interact with our platform, their actions trigger HTML requests, and the system replies with HTML responses.

For enhanced visual appeal and functionality, our platform fetches external CSS and JavaScript files from dedicated static file storage, ensuring speedy resource loading. When geographical data retrieval, the system utilizes the GeoJSON file format for efficient and precise location searches.

Our data storage solution utilizes SQLite, a robust and reliable database engine. This choice allows the efficient management of data and ensures easy data access, which is crucial to the performance of our web application.

Powered by the Python Flask MicroServices framework, the Web Server Gateway Interface (WSGI) channels data requests and triggers corresponding functions. This makes our system modular and adaptive.

Central to our architecture is the Recommendation Model Service, which uses stored data and user preferences to generate tailored airline suggestions, emphasizing our commitment to personalizing the user journey.

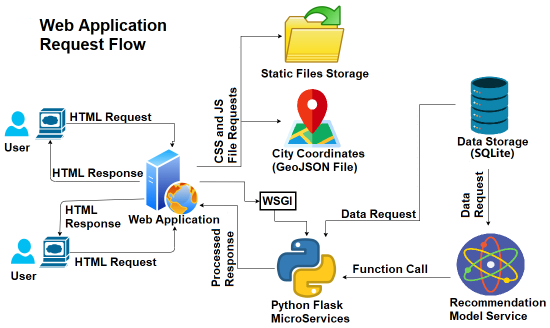


Figure 3: Web Application Architecture

4.5 Request Flow and Data Processing

The intricate relationship between user interactions and backend data processing is demonstrated in Figure 4. This relationship establishes the core of our web application's functionality, ensuring both a smooth user experience and efficient data handling.

A user starts the interaction by making a "City List Request" through the front end. This prompts the web application, built on the Python Flask framework, to send a "Database Request" to the data storage. At this stage, the SQLite facilitates easy and reliable data retrieval. In response, a list of cities gets generated and returned to the front end, appearing as a dropdown menu for the user.

As users indicate their preference for flight service, the app retrieves the information, passes it to our data model, and provides airline suggestions, which are shown on the interface for user clarity.

In the concluding feedback phase, users can view specific airline ratings, pulled from the data storage. This step enriches the user's understanding of available airline choices, facilitating informed decisions.

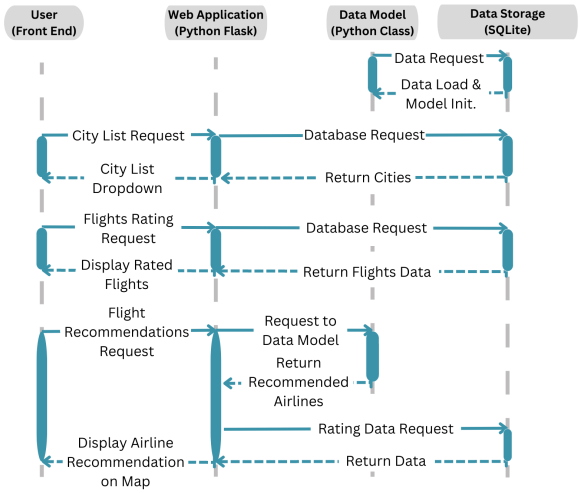


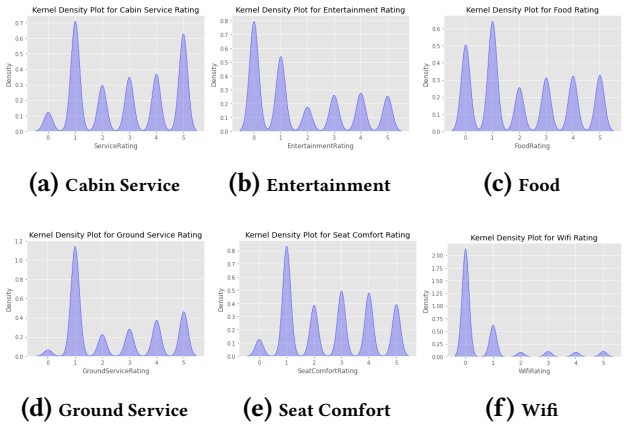
Figure 4: Request Flow & Data Process

5 EXPERIMENTS AND EVALUATION

5.1 Exploratory Data Analysis

Our exploratory data analysis on a dataset with 195,402 data points highlighted key patterns in flight service ratings. Using Kernel Density Plots, shown in Figure 5, we visualized diverse passenger sentiments across services such as cabin service, entertainment, seat comfort, and others. We also examined the correlations between

service factors, shown in Figure 6, with some links between seat comfort and overall service satisfaction. Conversely, the Wifi Rating largely operated independently of other factors. Overall, the initial exploratory data analysis offers airlines a concise overview of passenger preferences and areas for potential enhancement.



(a) Cabin Service (b) Entertainment (c) Food

(d) Ground Service (e) Seat Comfort (f) Wifi

Figure 5: Service Rating Kernel Density Plots

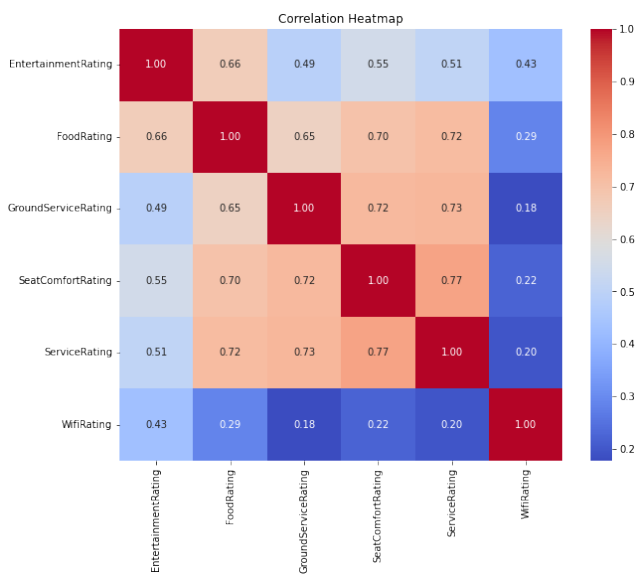


Figure 6: Correlation Heatmap

5.2 In-depth Analysis of Model Performances

To determine the best model for our recommendation system, multiple algorithms were evaluated using various configurations, as summarized in Table 1. This table showcases the performance of four algorithms: Random Forest (RF), K-Nearest Neighbors (KNN), Logistic Regression (LR), and RBF Kernel Logistic Regression (RBF LR). A total of four experiments using different segments of authentic trip data are presented in the table. Each experiment is marked with a unique number, and the optimal hyperparameters (HP) used for each model are provided.

No.	Model	Optimal HP	10-CV	Test
1	RF	t. num=29	0.8366 (OOB)	0.8769
	KNN	k=1	0.7717	0.6769
	LR	c=100	0.8234	0.8769
	RBF LR	g=0.1, c=100	0.6965	0.6923
2	RF	t. num=18	0.5478 (OOB)	0.6500
	KNN	k=17	0.5887	0.6250
	LR	c=50	0.5887	0.6500
	RBF LR	g=0.1, c=10	0.5650	0.6000
3	RF	t. num=23	0.5357 (OOB)	0.5714
	KNN	k=1	0.5214	0.5714
	LR	c=1	0.4571	0.5357
	RBF LR	g=0.1, c=100	0.5071	0.6429
4	RF	t. num=14	0.6535 (OOB)	0.8125
	KNN	k=1	0.6563	0.6563
	LR	c=1000	0.4976	0.4688
	RBF LR	g=0.1, c=1000	0.5713	0.5938

Table 1: Model Accuracy Comparison

Analyzing the data, we observe that the RF model consistently shows high accuracy, especially in experiments 1 and 4, compared to other models. The 10-Cross Validation (10-CV) scores provide insights into each model's stability and potential for generalization. Notably, the RF model exhibits the use of Out-of-Bag (OOB) estimation, an intrinsic form of validation that complements its predictive strength.

Experiment No.	1	2	3	4
% correctly labeled	0.9846	0.8750	0.7500	0.9688

Table 2: Percentage of Correct Labels Included in Top-3 Recommendations by Random Forest

Table 2 specifically highlights the proficiency of the Random Forest model in accurately including the real label within its top 3 recommended results across various experiments. This is essential because a high accuracy

in model prediction directly translates to the reliability of the airline recommendations made to users. The table demonstrates the RF model's consistency, with experiments 1 and 4 yielding notably high percentages of correct predictions, reinforcing the model's reliability.

In summary, these evaluation metrics emphasize the Random Forest's potential as a front-runner for our recommendation system. Its robustness, consistency, and predictive power underscore its suitability for the project, guiding users toward tailored airline selections based on empirical data.

5.3 Visualization Evaluation

For our project's success, user comprehension and satisfaction with our visual interfaces are paramount. To assess this, we invited a diverse group of participants to navigate our application as users and provide feedback. The survey covers various aspects of the visualization, including ease of understanding, ease of use, visual appeal, airline recommendation, likeliness to use in the future, and likeliness to recommend to others. Each participant needs to provide a rating for each aspect in the range of 1 to 5, with 5 being the highest score and 1 being the lowest score.

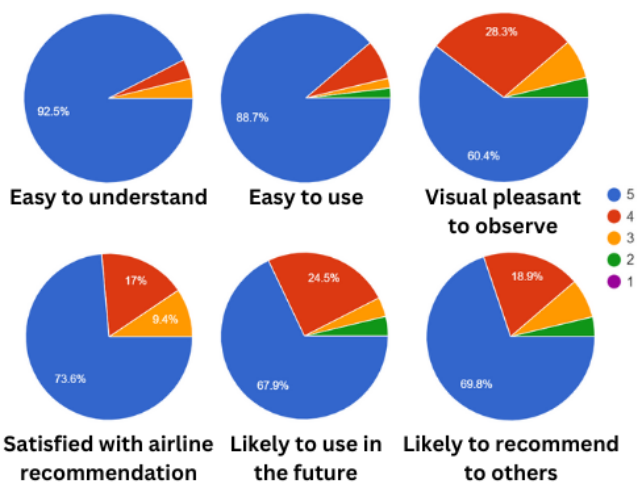


Figure 7: Visulization Evaluation Results

A total of 53 participants completed the survey, providing valuable feedback for this project. The results of the user feedback are shown in Figure 7. In evaluating the user assessment, ratings of 4 and 5 were considered approvals of the web application. A significant 96.3% of participants found the application easy to understand, while 96.2% agreed that the system was easy to use. There is more room for improvement regarding visual appeal, as the survey indicates that about 88.7% of users found it pleasant. The majority, 90.6%, expressed satisfaction with the airline recommendation results. Furthermore, 92.4% of participants indicated that they would likely use this system in the future, and 88.7% would likely recommend it to others.

Participants also provided insightful textual feedback with comments and suggestions for future improvement. The majority mentioned that they had never seen an app like this project before while some pointed out that the visualization could be further improved in future development. Additionally, some feedback suggested including a feature to verify authentic user reviews before adding them to the database, an area for future exploration beyond this project's current scope.

6 CONCLUSIONS & DISCUSSION

All team members have contributed a similar amount of effort to this project.

Our web application has innovated airline selection by personalizing recommendations using service-quality feedback, leveraging a Random Forest algorithm paired with dynamic visualization. However, challenges like scalability with vast review data suggest Logistic Regression might be more efficient in future expansions.

Subjectivity in reviews and the current model's focus on service experience, without pricing and flight specifics due to data constraints, are areas for future refinement. Additionally, enhancing the system to incorporate real-time user reviews and more accurate flight route simulations using real-world flight route data would improve the model's utility.

Despite these areas for growth, the project significantly impacts how users can choose an airline for their travel plan based on individual service preferences. It underscores the importance of user-centric design in recommendation systems that enrich user choice and planning process.

In summary, this project illustrates the transformative potential of combining user feedback with advanced data analytics, setting a path for continuous innovation in the travel industry. This system also marks a significant step toward a dynamic, responsive intuitive visualization during travel planning, with ample scope for future refinement and expansion.

REFERENCES

[1] Himan Abdollahpouri, Masoud Mansoury, Robin Burke, Bamshad Mobasher, and Edward Malthouse. 2021. User-centered Evaluation of Popularity Bias in Recommender Systems. UMAP '21: Proceedings of the 29th ACM Conference on User Modeling, Adaptation and Personalization. https://doi.org/10.1145/3450613.3456821

[2] Ebunoluwa Ashley-Dejo, Seleman Ngwira, and Tranos Zuva. 2016. A context-aware proactive recommender system for tourist. In *2016 International Conference on Advances in Computing and Communication Engineering (ICACCE)*. 271–275. https://doi.org/10.1109/ICACCE.2016.8073760

[3] Jesus Bobadilla, Fernando Ortega, Antonio Hernando, and Abraham Gutiérrez. 2013. Recommender systems survey. *Knowledge-Based Systems* 46 (2013), 109–132. https://doi.org/10.1016/j.knosys.2013.03.012

[4] Paul Brendel, Aracelis Torres, and Onyebuchi A Arah. 2023. Simultaneous adjustment of uncontrolled confounding, selection bias and misclassification in multiple-bias modelling. *International Journal of Epidemiology* 52 (2023), 1220–1230. Issue 4. https://doi.org/10.1093/ije/dyad001

[5] Jian Cao, Yuchang Xu, Huisi Ou, Yudong Tan, and Quanwu Xiao. 2018. PFS: A Personalized Flight Recommendation Service via Cross-Domain Triadic Factorization. In *2018 IEEE International Conference on Web Services (ICWS)*. 249–256. https://doi.org/10.1109/ICWS.2018.00039

[6] Amine Dadoun, Michael Defoin-Platel, Thomas Fiig, Corinne Landra, and Raphael Troncy. 2021. How recommender systems can transform airline offer construction and retailing. *Journal of Revenue and Pricing Management* 20 (2021), 301 – 315. https://doi.org/10.1057/s41272-021-00313-2

[7] R. De Maesschalck, Delphine Jouan-Rimbaud Bouveresse, and DésiréL Luc Luc Massart. 2000. The Mahalanobis distance. *Chemometrics and Intelligent Laboratory Systems* 50, 1 (2000), 1–18. https://doi.org/10.1016/S0169-7439(99)00047-7

[8] Colin Eberhardt. 2020. Rendering One Million Datapoints with D3 and WebGL. Scott Logic. https://blog.scottlogic.com/2020/05/01/rendering-one-million-points-with-d3.html Accessed: 10-07-2023.

[9] Antin Harasymiv. 2018. Prototyping a Smoother Map. Medium. https://medium.com/google-design/google-maps-cb0326d165f5 Accessed: 10-07-2023.

[10] Maryam Heidari and Setareh Rafatirad. 2020. Using Transfer Learning Approach to Implement Convolutional Neural Network model to Recommend Airline Tickets by Using Online Reviews. In *2020 15th International Workshop on Semantic and Social Media Adaptation and Personalization (SMA*. 1–6. https://doi.org/10.1109/SMAP49528.2020.9248443

[11] Emmanouela E Manganari, Evangelos Mourelatos, and Efthalia Dimara. 2021. Beyond the Lexical Sense of Online Reviews: The Role of Emoticons and Consumer Experience. *Interacting with Computers* 32, 5-6 (03 2021), 475–489. https://doi.org/10.1093/iwc/iwab004 arXiv:https://academic.oup.com/iwc/article-pdf/32/5-6/475/38856678/iwab004.pdf

[12] Elijah Meeks. 2015. *D3.js in Action*. Manning Publications, Chapter 11. Big Data Visualization. https://livebook.manning.com/book/d3-js-in-action/chapter-11/1

[13] Michael Siering, Amit Deokar, and Christian Janze. 2018. Disentangling consumer recommendations: Explaining and predicting airline recommendations based on online reviews. *Decision Support Systems* 107 (2018), 52–63. https://doi.org/10.1016/j.dss.2018.01.002

[14] Mona Taghavi, Jamal Bentahar, Kaveh Bakhtiyari, and Chihab Hanachi. 2017. New Insights Towards Developing Recommender Systems. *Comput. J.* 61, 3 (06 2017), 319–348. https://doi.org/10.1093/comjnl/bxx056

[15] Saravanan Thirumuruganathan, Soon gyo Jung, Dianne Ramirez Robillos, Joni Salminen, and Bernard Jansen. 2021. Forecasting the nearly unforecastable: why aren't airline bookings adhering to the prediction algorithm? *Electronic Commerce Research* 21 (2021), 73–100. https://doi.org/10.1007/s10660-021-09457-0

[16] Weiran Wang and Miguel Á. Carreira-Perpiñán. 2014. The role of dimensionality reduction in linear classification. *CoRR* abs/1405.6444 (2014). arXiv:1405.6444 http://arxiv.org/abs/1405.6444